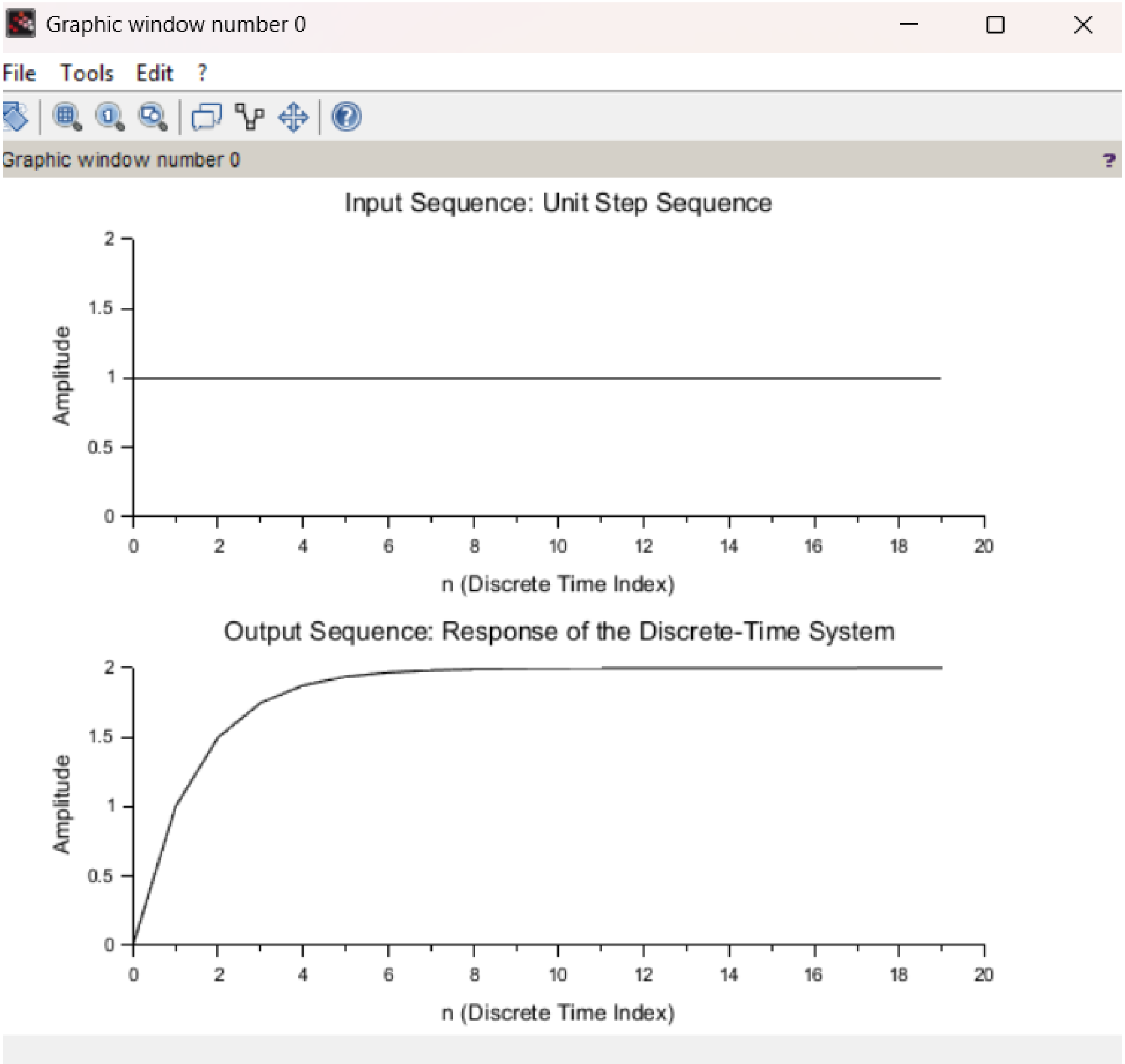
4. SYSTEM ANALYSIS AND CLASSIFICATION

Experiment 4.1

```
// Test Case: Analysis of a Discrete-Time Linear System

// Define parameters
N = 20; // Number of samples
n = 0:N-1; // Discrete time index vector
x_n = ones(1, N); // Input sequence: Unit step sequence
y_n = zeros(1, N); // Output sequence initialization
// Define the difference equation: y[n] = 0.5y[n-1] + x[n]
for k = 2:N
y_n(k) = 0.5 * y_n(k-1) + x_n(k);
end
// Plot the input sequence
clf; // Clear the current figure
subplot(2,1,1);
plot2d (n, x_n);
xlabel('n (Discrete Time Index)');
ylabel('Amplitude');
title('Input Sequence: Unit Step Sequence');
// Plot the output sequence
subplot(2,1,2);
```

```
plot2d(n, y_n);
xlabel('n (Discrete Time Index)');
ylabel('Amplitude');
title('Output Sequence: Response of the Discrete-Time System');
```



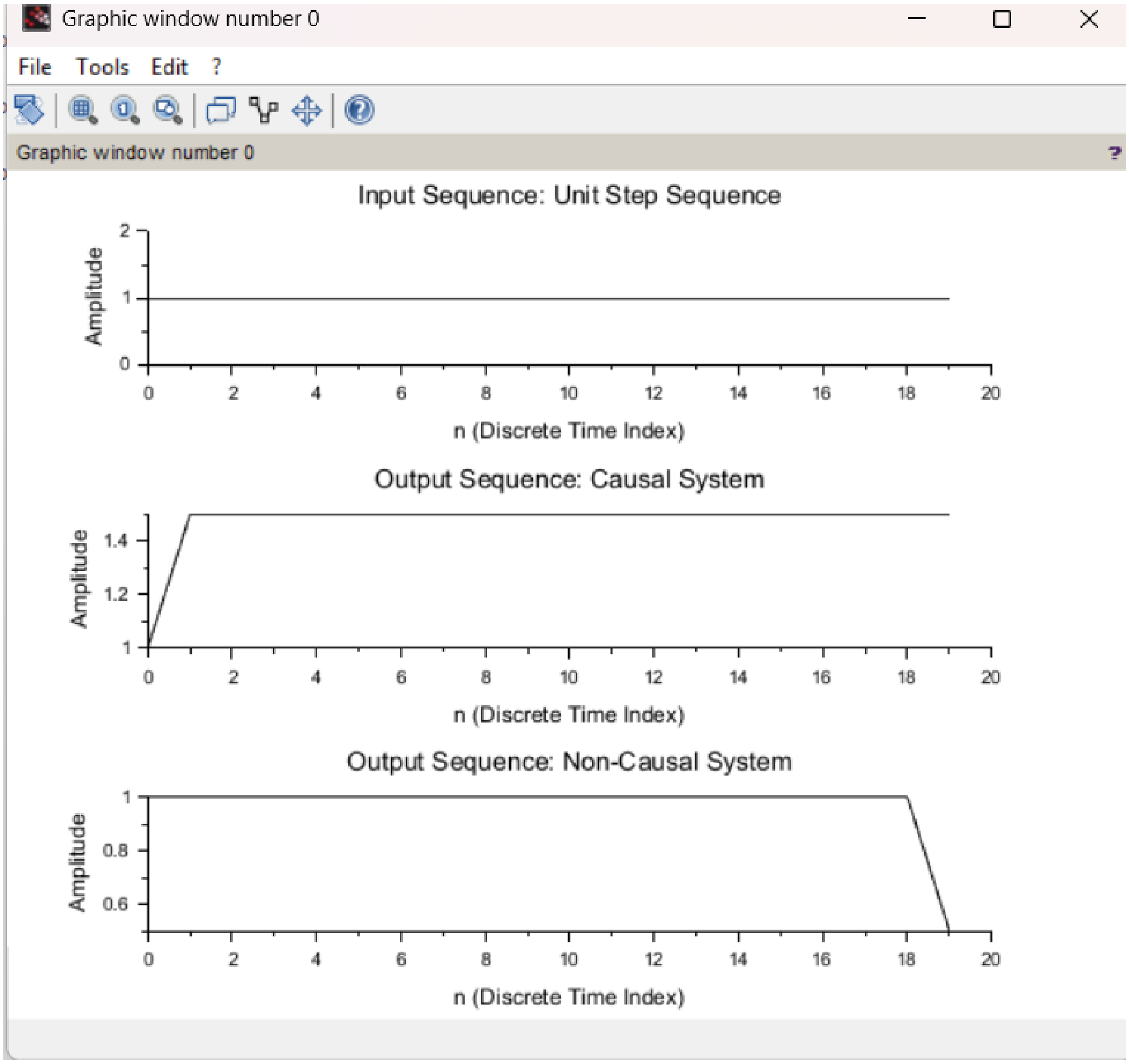
Experiment 4.2

```
// Test Case: Causal and Non-Causal Systems Analysis
// Define parameters
N = 20; // Number of samples
n = 0:N-1; // Discrete time index vector
x_n = ones(1, N); // Input sequence: Unit step sequence
// Initialize output sequences
y_causal = zeros(1, N); // Output sequence for causal system
y_noncausal = zeros(1, N); // Output sequence for non-causal
system
// Causal System: y[n] = x[n] + 0.5x[n-1]
for k = 1:N
if k > 1 then
y_causal(k) = x_n(k) + 0.5 * x_n(k-1);
else
y_causal(k) = x_n(k); // No x[n-1] for k=0
end
end

// Non-Causal System: y[n] = 0.5x[n] + 0.5x[n+1]
for k = 1:N
if k < N then
```

```
y_noncausal(k) = 0.5 * x_n(k) + 0.5 * x_n(k+1);
else
y_noncausal(k) = 0.5 * x_n(k); // No x[n+1] for k=N-1
end
end
// Plot the input sequence
clf; // Clear the current figure
subplot(3,1,1);
plot2d(n, x_n);
xlabel('n (Discrete Time Index)');
ylabel('Amplitude');
title('Input Sequence: Unit Step Sequence');
// Plot the output of the causal system
subplot(3,1,2);
plot2d(n, y_causal);
xlabel('n (Discrete Time Index)');
ylabel('Amplitude');
title('Output Sequence: Causal System');
// Plot the output of the non-causal system
subplot(3,1,3);
plot2d(n, y_noncausal);
xlabel('n (Discrete Time Index)');
```

ylabel('Amplitude');

title('Output Sequence: Non-Causal System');



Experiment 4.3

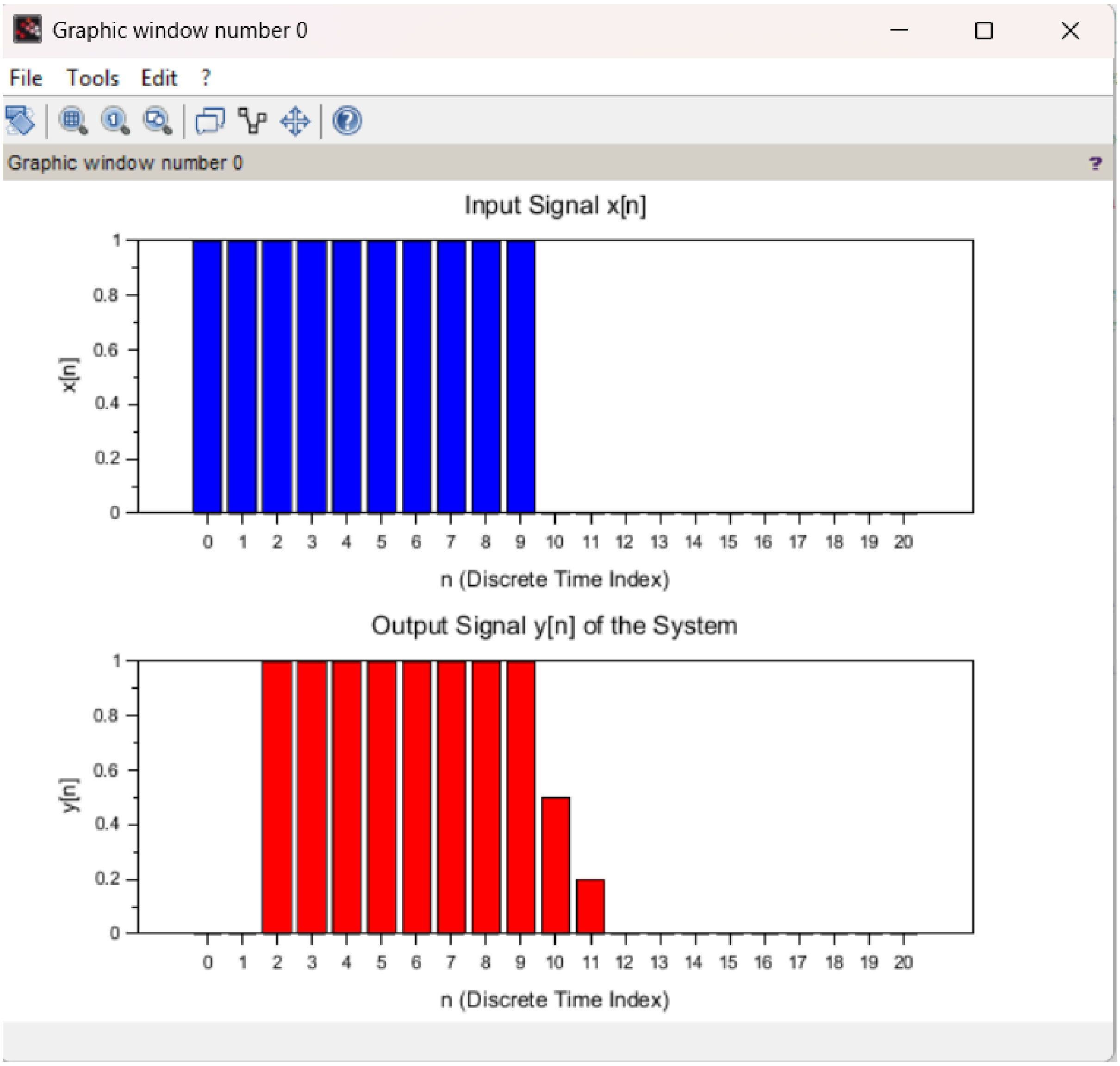
// Test Case: Analysis of a Discrete-Time System

// Define the input signal x[n]

n = 0:20; // Discrete time index

x_n = (n >= 0) - (n >= 10); // Example input: A rectangular pulse from n=0 to n=9

```
// Initialize the output signal y[n] with zeros
y_n = zeros(1, length(n));
// Define the system coefficients
b = [0.5, 0.3, 0.2]; // Coefficients of x[n], x[n-1], and x[n-2]
// Compute the output y[n] using the difference equation
for i = 3:length(n)
y_n(i) = b(1) * x_n(i) + b(2) * x_n(i-1) + b(3) * x_n(i-2);
end

// Plot the input signal x[n]
clf; // Clear the current figure
subplot(2,1,1);
bar(n, x_n, 'b');
xlabel('n (Discrete Time Index)');
ylabel('x[n]');
title('Input Signal x[n]');
// Plot the output signal y[n]
subplot(2,1,2);
bar(n, y_n, 'r');
xlabel('n (Discrete Time Index)');
ylabel('y[n]');
title('Output Signal y[n] of the System');
```



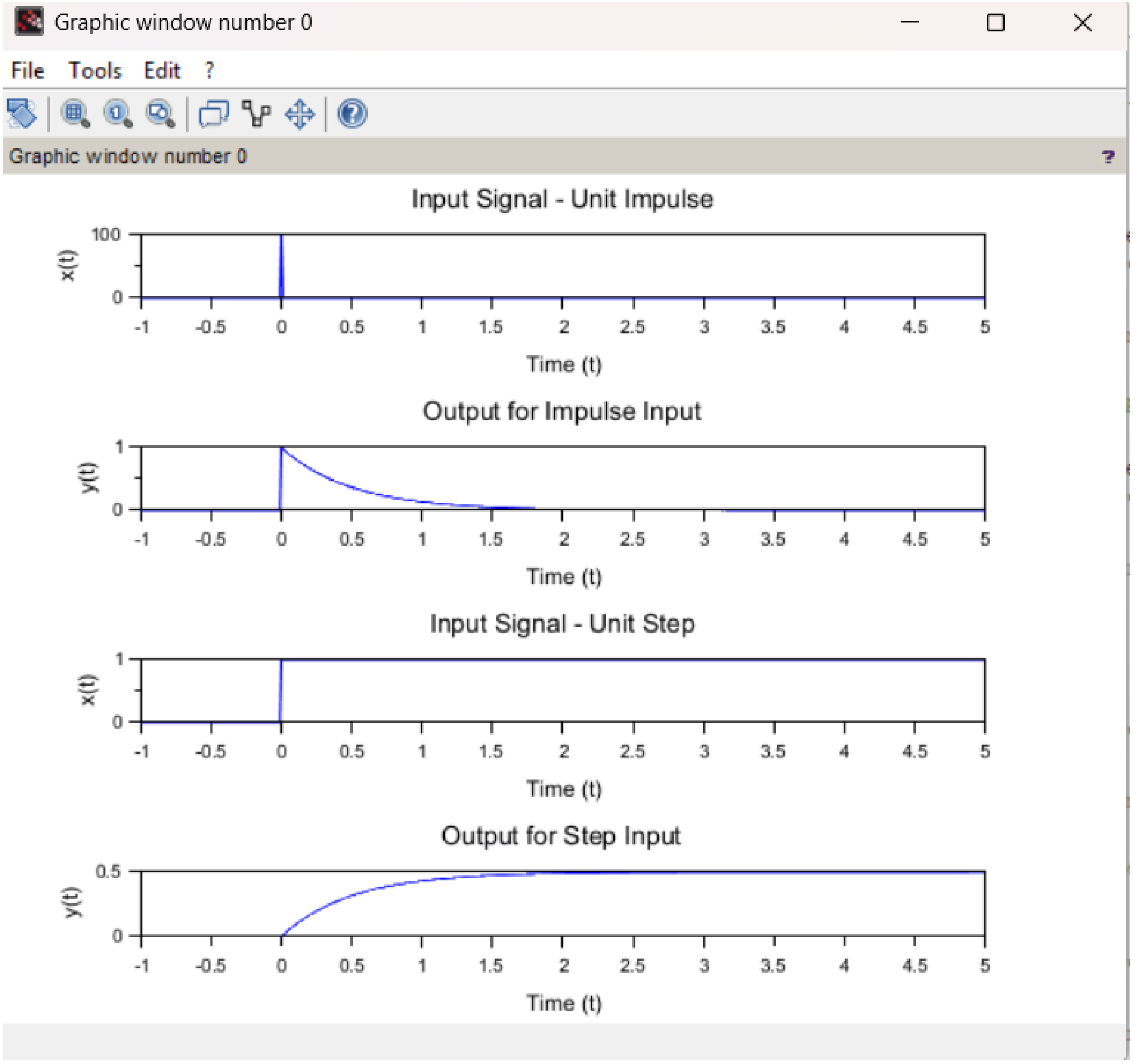
Experiment 4.4

```
clc;

clear;

clf;


//-----------------------------------

// Analysis of a Continuous-Time System

//-----------------------------------
```

```
// Time range
t_min = -1;
t_max = 5;
dt = 0.01;
t = t_min:dt:t_max;

// Input Signals

// Impulse (Approximation)
x_impulse = zeros(t);
index = find(abs(t) < dt/2);
x_impulse(index) = 1/dt;

// Unit Step
x_step = zeros(t);
x_step(t >= 0) = 1;

// System Parameters
a = 2;
b = 1;
```

```
// Output for Impulse Input
// h(t) = (b/a)e^(-at)u(t)
y_impulse = zeros(t);
y_impulse(t >= 0) = b * exp(-a * t(t >= 0));

// Output for Step Input
// y(t) = (b/a)(1-e^(-at))u(t)
y_step = zeros(t);
y_step(t >= 0) = (b/a) * (1 - exp(-a * t(t >= 0)));

//-------------------------------------
// Plot Results
//-------------------------------------

// Input Impulse
subplot(4,1,1);
plot(t, x_impulse);
xlabel("Time (t)");
ylabel("x(t)");
title("Input Signal - Unit Impulse");

// Output for Impulse
```

```
subplot(4,1,2);
plot(t, y_impulse);
xlabel("Time (t)");
ylabel("y(t)");
title("Output for Impulse Input");

// Input Step
subplot(4,1,3);
plot(t, x_step);
xlabel("Time (t)");
ylabel("x(t)");
title("Input Signal - Unit Step");

// Output for Step
subplot(4,1,4);
plot(t, y_step);
xlabel("Time (t)");
ylabel("y(t)");
title("Output for Step Input");
```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File Tools Edit ?
Graphic window number 0
Input Signal - Unit Impulse
x(t)
100
0
-1 -0.5 0 0.5 1 1.5 2 2.5 3 3.5 4 4.5 5
Time (t)
Output for Impulse Input
y(t)
1
0
-1 -0.5 0 0.5 1 1.5 2 2.5 3 3.5 4 4.5 5
Time (t)
Input Signal - Unit Step
x(t)
1
0
-1 -0.5 0 0.5 1 1.5 2 2.5 3 3.5 4 4.5 5
Time (t)
Output for Step Input
y(t)
0.5
0
-1 -0.5 0 0.5 1 1.5 2 2.5 3 3.5 4 4.5 5
Time (t)